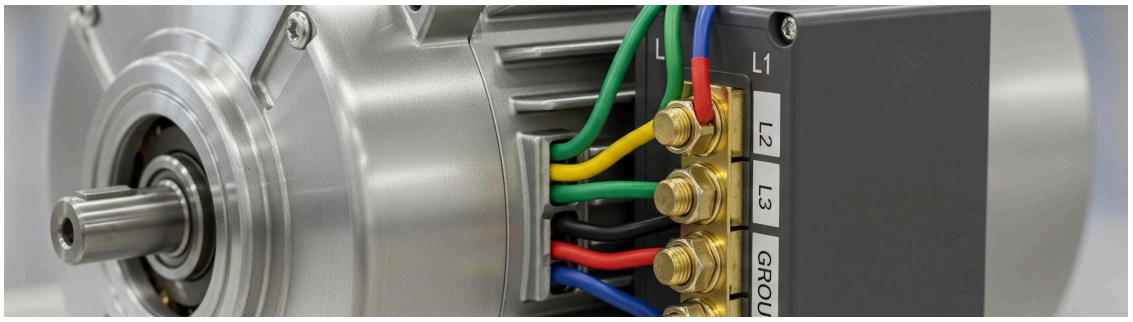

Electrical Machine Data Sheet

Machine Name/Type : DC Series motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

Parameter	Value	Unit
Power/Output	5	HP
Voltage	220	V
Current (Rated)	19	A
Frequency	NA	NA
Speed	1500	RPM

Precautions

I. Ensure the supply voltage is 220 V DC before starting the motor.

II . Never operate the motor at no-load. A DC series motor can overspeed dangerously.

III . Always use a starter to limit high starting current.

IV . Check all terminal connections as per the circuit diagram before switching ON.

V . Do not allow armature current to exceed the rated value of 25 A.

VI . Apply load before starting and increase it gradually.

VII . Ensure proper earthing of the motor body to avoid electric shock.

VIII . Keep hands and tools away from rotating parts during operation.

IX . Continuously monitor speed, noise, and temperature.

X . Switch OFF immediately if abnormal vibration, sparking, or overheating occurs.

Working Principle of DC Series Motor

A DC series motor works on the principle that a **current-carrying conductor placed in a magnetic field experiences a force**.

In this motor, the **field winding is connected in series with the armature**, so the same current flows through both.
When supply is given, a strong magnetic field is produced, and interaction between armature current and magnetic flux generates **high starting torque**.

Working Characteristics of DC Series Motor

I. Produces very high starting torque.

II. Speed varies significantly with load.

III. Speed increases rapidly at light or no load (unsafe condition).

IV. Torque is approximately proportional to the square of armature current at low saturation.

V. Poor speed regulation compared to shunt motor.

Applications

1. Electric traction (trains, trams)
2. Cranes and hoists
3. Elevators
4. Rolling mills
5. Electric locomotives
6. Winches
7. Conveyors with heavy loads
8. Automobile starter motors
9. Heavy-duty compressors
10. Industrial lifting equipment